Ribonuclease E (rne / PP_1905, UniProt Q88LM4) in *Pseudomonas putida* KT2440: A Functional Annotation Report

UniProt: Q88LM4 · **Gene:** *rne* · **Locus:** PP_1905 · **EC:** 3.1.26.12 **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950 / KT2440)

Summary

The **rne** gene of *Pseudomonas putida* KT2440 (UniProt **Q88LM4**, ordered locus **PP_1905**) encodes **Ribonuclease E (RNase E; EC 3.1.26.12)**, a **1,091-residue (≈120.9 kDa), Mg^{2+}-dependent, single-strand-specific endoribonuclease** that is the central enzyme of bacterial RNA metabolism. Its primary catalytic function is the **endonucleolytic cleavage of single-stranded RNA in A- and U-rich regions**. It performs two overarching biological tasks: (1) the **maturation of stable RNA** — the 5′ processing of 16S and 5S ribosomal RNAs and the majority of tRNAs — and (2) the **initiation of messenger RNA decay**, in which its cleavage of the monophosphorylated intermediate is typically the rate-limiting, committed step of mRNA turnover.

A critical identification issue was resolved during this investigation. Although one automated UniProt annotation tag lists the "RNase G subfamily," the protein's length and domain architecture identify it **unambiguously as full-length RNase E**, not the shorter RNase G paralog. Q88LM4 comprises an N-terminal catalytic half (S1 RNA-binding domain, Mg^{2+} active site, and a zinc-linked tetramerization region) followed by a very large **C-terminal intrinsically disordered region (~residues 506–1091)** that is entirely absent from RNase G (~489 aa). This C-terminal domain (CTD) is the **scaffold of the RNA degradosome**, a multi-enzyme machine that recruits the exoribonuclease PNPase, the DEAD-box RNA helicase RhlB, and (in *E. coli*) enolase. The organism additionally encodes a *separate* RNase G (rng), confirming that PP_1905 is the RNase E ortholog.

Functionally, the enzyme carries out its work on the **cytoplasmic face of the inner membrane**, to which it is tethered as a peripheral membrane protein via an amphipathic helix in its CTD. This localization spatially organizes RNA turnover within the cell. Substrate

recognition is governed by a dedicated **5′-monophosphate sensor pocket** and a **5′→3′ linear scanning mechanism** whose progress is impeded by RNA secondary structure, bound ribosomes, and base-paired small RNAs — a mechanism that sets mRNA lifetimes. Direct experimental evidence in the exact target organism (*P. putida* KT2440) demonstrates that RNase E is a non-redundant determinant of environmental lifestyle and stress endurance, with species-specific physiological roles that diverge from *E. coli*.

Key Findings

Finding 1 — Q88LM4 is full-length RNase E, not RNase G

The single most important interpretive result of this investigation is the correct sub-family assignment. UniProt Q88LM4 is **1,091 amino acids / 120.9 kDa**. Its domain architecture consists of: an N-terminal **S1 RNA-binding motif** (residues ~39–117); a **catalytic core** with Mg^{2+} active-site binding residues (positions ~300 and 343); a **zinc-tetramerization region** (residues 401–404) annotated as "required for zinc-mediated homotetramerization and catalytic activity"; and a **very large C-terminal intrinsically disordered region** (~506–1091) containing multiple low-complexity and charged segments.

This architecture matches *E. coli* RNase E (~1,061 aa; a catalytic N-domain plus a C-terminal scaffold) and is fundamentally distinct from the short RNase G paralog (~489 aa; catalytic domain only, no C-terminal scaffold). The authoritative **HAMAP-Rule MF_00970** assigns the "RNase E subfamily"; a conflicting, lower-quality automated ARBA rule listing "RNase G subfamily" is an annotation artifact superseded by the unambiguous full-length architecture. Importantly, *P. putida* KT2440 encodes a **separate RNase G** (rng), which was studied independently (see Finding 8), removing any doubt that PP_1905 is the RNase E ortholog.

Finding 2 — Primary function: Mg^{2+}-dependent, single-strand-specific endoribonuclease cleaving A/U-rich RNA (EC 3.1.26.12)

The catalytic activity, per UniProt/HAMAP MF_00970, is: **"Endonucleolytic cleavage of single-stranded RNA in A- and U-rich regions"** (EC 3.1.26.12), with a cofactor requirement of **1 Mg^{2+} per subunit** at the active site. This is corroborated by mechanistic structural biology: Koslover et al. (2008) describe RNase E as *"an essential bacterial endoribonuclease involved in the turnover of messenger RNA and the maturation of structured RNA precursors in Escherichia coli"* and report a crystal structure revealing *"a mechanism of RNA recognition and cleavage that* *explains the enzyme's preference for substrates possessing a 5′-monophosphate"* (PMID: 18682225). The dual role — mRNA turnover plus structured-RNA maturation — is the defining functional signature of RNase E.

Finding 3 — 5′-monophosphate sensing and 5′→3′ scanning govern substrate access and cleavage-site selection

RNase E does not cleave RNA at random. It preferentially binds a **5′-monophosphorylated end** via a dedicated 5′-sensor pocket, which potentiates catalytic efficacy (Hui & Belasco 2021, PMID: 34643703; Jourdan & McDowall 2008). Richards & Belasco (2019) demonstrated that the enzyme *"searches for cleavage sites by scanning linearly from the 5′-terminal monophosphate along single-stranded regions of RNA and that its progress is impeded by structural discontinuities"* (PMID: 30852060). These discontinuities — bound proteins, translating ribosomes, and small-RNA base-pairing — are the physical determinants that set individual mRNA lifetimes.

A second, **5′-end-independent "direct entry"** mode also exists, in which RNase E *"can cleave certain RNAs rapidly without requiring a 5′-monophosphorylated end"* by simultaneously engaging multiple single-stranded segments (Kime et al. 2010, PMID: 19889093). The 5′-monophosphate sensing property is conserved across the RNase E/G family: for RNase G, *"the presence of a 5′ monophosphate can enhance the affinity of RNase G binding to RNA"* (PMID: 18078441).

Finding 4 — Quaternary structure: a zinc-linked homotetramer (dimer of dimers)

The catalytic domain assembles into a **homotetramer** whose integrity depends on a shared Zn^{2+} ion — the so-called **"Zn-link."** Callaghan et al. (2005) showed that *"two protomers share a single zinc ion"* and that *"catalytic activity does not require zinc directly but does require the quaternary structure, for which the metal is essential"* (PMID: 15779893). Mutation of the Zn-coordinating residues causes partial zinc loss, disruption of the tetramer into dimers, and effective catalytic inactivation.

Q88LM4 preserves this architecture exactly: residues 401–404 are annotated as "required for zinc-mediated homotetramerization and catalytic activity," and UniProt records 2 Zn^{2+} per homotetramer with a "dimer of dimers" subunit state. Interestingly, a minimal ~395-residue catalytic peptide can retain scanning and cleavage even without the Zn-link (Caruthers et al.

2006, PMID: 16854990), which delimits the essential catalytic core and shows that the tetrameric quaternary structure, while physiologically important, is not strictly required for core enzymatic activity.

Finding 5 — Biological processes: 5′-maturation of 16S/5S rRNA and most tRNAs, plus initiation of mRNA decay

HAMAP MF_00970 assigns to Q88LM4: *"Required for the maturation of 5S and 16S rRNAs and the majority of tRNAs. Also involved in the degradation of most mRNAs."* Each arm is supported by direct experimental work in *E. coli*:

- **rRNA maturation:** RNase E and RNase G together perform two-step 5′ maturation of 16S rRNA. Li et al. (1999) showed that *"when both RNase E and CafA are inactivated, 5′ maturation of 16S rRNA is completely blocked"* (PMID: 10329633).
- **tRNA maturation:** RNase E generates the mature 3′ termini of the proline tRNAs by a single endonucleolytic cut immediately after the CCA determinant. Mohanty et al. (2016) reported that *"RNase E is primarily responsible for the endonucleolytic removal of the entire Rho-independent transcription terminator associated with the proK, proL and proM primary transcripts by cleaving immediately downstream of the CCA determinant"* (PMID: 27288443).
- **mRNA decay:** RNase E cleavage of the monophosphorylated intermediate is frequently the committed, rate-limiting step of turnover. Luciano et al. (2012) found that for such transcripts *"their decay rate is limited by cleavage of the monophosphorylated intermediate, making RNase E critical for their rapid turnover"* (PMID: 22984254).

Finding 6 — Degradosome scaffold: the C-terminal disordered domain nucleates a multi-enzyme RNA-degradation machine

The UniProt SUBUNIT annotation for Q88LM4 states that it is a *"Component of the RNA degradosome, which is a multiprotein complex involved in RNA processing and mRNA degradation."* In *E. coli*, the degradosome is *"typically composed of the endoribonuclease RNase E, which also serves as a scaffold for the other components, the exoribonuclease PNPase, the RNA helicase RhlB, and enolase"* (Regonesi et al. 2006, PMID: 16139413).

Critically, the scaffolding role maps to the **RNase E C-terminal half**. Domínguez-Malfavón et al. (2013) found that degradosome interactions are *"absent in a mutant strain that lacks the C-terminal half, supporting the role of the carboxy-end domain as the scaffold for the degradosome"* (PMID: 23927922). Q88LM4 carries exactly this large C-terminal intrinsically disordered region (~506–1091), providing a direct structural basis for its scaffolding role.

Finding 7 — Localization: cytoplasmic face of the inner membrane, spatially organizing mRNA degradation

The UniProt subcellular location for Q88LM4 is **Cytoplasm and Cell inner membrane (peripheral, cytoplasmic side)**. In *E. coli*, RNase E binds the inner membrane through an amphipathic "Segment A" helix in its CTD, and this tethering shapes where decay occurs. Kim et al. (2026) reported that co-transcriptional mRNA degradation *"is rare due to membrane localization of RNase E, except for transcripts encoding inner-membrane proteins"* (PMID: 42174289). Single-molecule imaging confirms that the membrane-binding motif and the C-terminal domain govern RNase E localization and diffusion (Troyer et al. 2025, PMID: 40093181).

Finding 8 — Direct *P. putida* KT2440 evidence: RNase E is a distinct, non-redundant determinant of lifestyle and stress endurance

The most directly relevant experimental evidence comes from the exact target organism. Apura et al. (2021) constructed single-RNase deletion variants of *P. putida* KT2440 (PNPase, RNase R, RNase E, RNase III, RNase G) and analyzed growth, motility, morphology, and oxidative-stress responses, concluding that *"each ribonuclease is specifically related with different traits of the environmental lifestyle that distinctively characterizes this microorganism"* (PMID: 33089610). Critically, *"the physiological responses of P. putida to the absence of each enzyme diverged significantly from those known previously in Escherichia coli,"* revealing species-specific regulatory functions. This study confirms in vivo that RNase E (PP_1905) functions in post-transcriptional adaptation and is functionally distinguishable from its paralog RNase G (rng).

Finding 9 — Pseudomonas degradosome assembles on the RNase E CTD via short linear motifs; autoregulation tunes enzyme level

In Pseudomonadota, the degradosome *"assembles on the C-terminal domain (CTD) of RNase E through short linear motifs (SLiMs) that determine its composition and functionality"* (Geslain et al. 2025, PMID: 40096066). In the closely related *P. aeruginosa*, the CTD's SLiMs mediate membrane attachment, RNA binding, complex clustering, and direct binding to PNPase and RhlB; CTD mutants are impaired in cold adaptation, pH response, and virulence.

The RNase E–PNPase interaction is conserved but evolutionarily re-wired: Paris et al. (2025) found that *"a different recognition mode arose for Pseudomonas aeruginosa, illustrating the evolutionary drive to maintain physical association of the two ribonucleases"* (PMID: 41036625).

Enzyme level is feedback-controlled: RNase E *"controls the decay of its own mRNA by cleaving it within the 5'-untranslated region (UTR), thereby autoregulating its synthesis"* (Sousa et al. 2001, PMID: 11722748), via an evolutionarily conserved 5′UTR stem-loop sensor (Diwa et al. 2000, PMID: 10817759). Notably, *P. aeruginosa* RhlB is a distinct "Type II" helicase that RNase E regulates by antagonizing its phase separation rather than allosterically activating it (Hausmann et al. 2026, PMID: 42581758).

Finding 10 — Sequence-level confirmation: N-terminus and Zn-link CxxC signature are diagnostic of RNase E

Direct analysis of the Q88LM4 sequence (1,091 aa) seals the identification. The N-terminus is **'MKRMLINATQPEELRVALVDGQRLYDL...'**, essentially identical to the diagnostic N-terminal signature of *E. coli* RNase E ('MKRMLINATQ...') and distinct from RNase G. The **Zn-link motif is intact**: cysteines occur precisely at positions **401 and 404 (a CxxC pair)**, matching the UniProt-annotated "zinc-mediated homotetramerization and catalytic activity" region (401–404). Compositional analysis confirms the two-domain architecture: the N-terminal catalytic half (res 1–510) has a globular-like charged fraction (DEKR ≈ 0.30), whereas the C-terminal half (511–1091) is strongly enriched in charged residues (DEKR ≈ 0.38) and disorder-promoting residues (P/A/Q/S/G ≈ 0.44) — the compositional hallmark of the intrinsically disordered degradosome scaffold. RNase G (~489 aa) has no such C-terminal extension.

Mechanistic Model / Interpretation

The findings integrate into a coherent picture of a single, bifunctional, membrane-anchored enzyme that governs the flow of RNA through the bacterial cell.

Domain architecture (Q88LM4, 1,091 aa)

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N ──────────── CATALYTIC HALF (~1–510) ──────────── │ ── C-TERMINAL DISORDERED SCAFFOLD (~506–1
  |        |           |          |                  |
 S1 RNA   Mg2+       Zn-link     5'-sensor           SLiMs: membrane helix, RNA-binding,
 domain   active   CxxC @401/404  pocket             PNPase-, RhlB-binding, clustering
(39–117)  site                                       (intrinsically disordered; DEKR~0.38)
         (~300,343)
  \______ homotetramer (dimer of dimers) ______/      \___ nucleates the RNA degradosome ___/
                (2 Zn2+ per tetramer)                        tethered to inner membrane
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Catalytic cycle and substrate selection

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5'-PPP-mRNA  ──(RppH pyrophosphohydrolase)──►  5'-P-mRNA
                                                    │
                                                    ▼
                                     RNase E 5'-sensor pocket binds 5'-monophosphate
                                                    │
                                            5'→3' linear scanning along ssRNA
                                                    │
                ┌───────────────────────────────────┴──────────────────────────────────────┐
                ▼                                                                            ▼
obstacle (ribosome, structure,                                    A/U-rich single-stranded site 
sRNA duplex) → scanning impeded                                                 │
→ longer mRNA lifetime                                                          ▼
                                                           Mg2+-dependent endonucleolytic cleavage
                                                                               │
                                                       ┌───────────────────────┴───────────┐
                                                       ▼                                     ▼
                                      downstream fragments handed to             upstream fragme
                                      PNPase/RhlB in the degradosome             (new monophosph
```

An alternative **"direct entry"** pathway bypasses the 5′-sensor, engaging multiple single-stranded segments at once, allowing rapid cleavage of certain substrates independent of the 5′-phosphorylation state.

The two jobs of RNase E

Function	Substrate	Outcome	Key evidence
Stable-RNA maturation	16S rRNA precursor	Mature 5′ end (with RNase G)	PMID: 10329633
	5S rRNA precursor	Mature 5′ end	HAMAP MF_00970
	Majority of tRNA precursors; proK/proL/proM	Mature 3′ end (cut after CCA)	PMID: 27288443
mRNA decay initiation	Most cellular mRNAs	Rate-limiting first cut → degradosome	PMID: 22984254
Autoregulation	Own rne mRNA 5′UTR	Feedback control of enzyme level	PMID: 11722748

Spatial organization

The enzyme is a **peripheral inner-membrane protein** (cytoplasmic side). Because the degradosome is membrane-tethered while transcription/translation proceed in the nucleoid/ cytoplasm, most mRNA degradation is spatially segregated from transcription — except for mRNAs encoding inner-membrane proteins, whose translation brings them to the membrane where RNase E resides. This membrane localization is a genuine regulatory layer, not an incidental property.

RNase E vs. RNase G — resolving the annotation conflict

Feature	RNase E (Q88LM4/ PP_1905)	RNase G (rng, separate gene)
Length	1,091 aa (~121 kDa)	~489 aa (~50 kDa)
N-terminal catalytic domain	Yes	Yes (35% identity to RNase E N-domain)
Zn-link CxxC (401/404)	Yes	Variable
C-terminal disordered scaffold	**Yes (~506–1091)**	**No**
Degradosome scaffold	**Yes**	No
Membrane tether	Yes	No
Essentiality	Essential	Non-essential

The full-length architecture, the diagnostic N-terminal 'MKRMLINATQ' signature, the intact Zn-link, and the presence of a separate rng gene in the genome collectively establish that PP_1905 is **RNase E**, and that the automated "RNase G subfamily" tag is an annotation artifact.

Evidence Base

PMID	Study (abbrev.)	How it supports the findings
18682225	Crystal structure of E. coli RNase E apoprotein	Enzyme class, dual role, and 5′-monophosphate preference (F2)
30852060	Obstacles to scanning govern mRNA lifetimes	Defines 5′→3′ linear scanning, obstacle-limited cleavage (F3)
19889093	Rapid cleavage without 5′ monophosphate	Documents the "direct entry" 5′-end-independent mode (F3)
18078441	Sensing of 5′ monophosphate by RNase G	5′-monophosphate sensing conserved in RNase E/G family (F3)
15779893	"Zn-link" metal-sharing interface	Zn-linked tetramer; quaternary structure essential (F4)
16854990	Minimal RNase E peptide lacking tetramer domain	Delimits minimal catalytic core (F4)
10329633	RNase G + RNase E in 16S rRNA 5′ maturation	Essential role in 16S rRNA maturation (F5)
27288443	RNase E generates mature 3′ termini of Pro tRNAs	Direct role in tRNA 3′-end maturation (F5)
22984254	Control of 5′-end-dependent mRNA degradation	RNase E as rate-limiting step in mRNA decay (F5)
16139413	Degradosome composition by proteomics	Names RNase E scaffold; lists PNPase, RhlB, enolase (F6)
23927922	Assembly/distribution of E. coli degradosome	Localizes scaffolding to C-terminal domain (F6)
42174289	Spatial constraints on mRNA degradation	Ties inner-membrane localization to decay pattern (F7)
40093181	Single-molecule imaging of RNase E	Membrane motif + CTD govern localization/diffusion (F7)
33089610	Ribonucleases control traits of P. putida	**Direct evidence in target organism** (F8)
40096066		

PMID	Study (abbrev.)	How it supports the findings
	RNase E scaffolding domain in P. aeruginosa	Pseudomonas degradosome assembles on CTD via SLiMs (F9)
41036625	Multi-dentate RNase E–PNPase interaction	Association conserved but re-wired in Pseudomonas (F9)
11722748	Autoregulation of E. coli RNase E	Autoregulation via 5′UTR self-cleavage (F9)
10817759	Conserved RNA stem-loop sensor for rne	Conserved 5′UTR stem-loop drives autoregulation (F9)
42581758	RNase E counteracts RhlB phase separation	Pseudomonas Type II RhlB regulation via condensate control (F9)

Supporting context papers: RNase G quaternary structure and dimerization (PMID: 14622423); RNase G in 16S/23S rRNA processing (PMID: 10362534, PMID: 21717341); functional divergence of RNase E vs. RNase G (PMID: 20507976); *P. aeruginosa* RNase E variant, iron piracy and virulence (PMID: 37027441); phage-encoded Dip inhibitor of the Pseudomonas degradosome (PMID: 27447594, PMID: 27834591); RraA modulator of RNase E (PMID: 21063756); RhlE helicase–RNase E interactions in Pseudomonas (PMID: 34151378, PMID: 38874491).

Note on the RNase E/G family nuance: Several papers (PMID: 20507976, PMID: 14622423) emphasize that RNase G and the RNase E catalytic domain share ~35% identity but have distinct in vivo activities — RNase G cannot substitute for RNase E under normal conditions. This underscores that even though Q88LM4's catalytic domain is homologous to RNase G, its full-length architecture and scaffold domain make it the functionally distinct RNase E.

Limitations and Knowledge Gaps

1. **Mechanistic evidence is largely extrapolated from *E. coli*.** The catalytic mechanism (5′-sensing, scanning, Zn-link tetramer, cleavage-site specificity) and the specific rRNA/tRNA maturation substrates are established in *E. coli*, not directly in *P. putida* KT2440. Apura et al. (2021, PMID: 33089610) explicitly caution that *P. putida* physiological responses diverge from *E. coli*, so extrapolation carries uncertainty.

2. **No enzymatic characterization of Q88LM4 itself.** There is no purified-protein biochemistry (kcat, Km, substrate profiling) reported specifically for the *P. putida* KT2440 RNase E. All catalytic parameters are inferred from orthologs.
3. **Degradosome composition in *P. putida* is inferred, not directly mapped.** The Pseudomonas degradosome has been characterized primarily in *P. aeruginosa* (PMID: 40096066, PMID: 41036625, PMID: 42581758). The exact SLiM inventory, partner set (e.g., which RhlB/RhlE helicase, whether enolase associates), and CTD interaction interfaces of the KT2440 protein remain to be experimentally defined.
4. **The specific mRNA regulon of *P. putida* RNase E is unknown.** Which transcripts are direct RNase E substrates, and how this shapes KT2440's environmental adaptation and biotechnological phenotypes (e.g., solvent tolerance, aromatic-compound catabolism), has not been mapped transcriptome-wide.
5. **Subcellular localization is annotation-based.** The inner-membrane, cytoplasmic-side localization for Q88LM4 comes from UniProt/ortholog inference; direct imaging in *P. putida* has not been performed.
6. **Essentiality and autoregulation of rne in KT2440** are assumed from the family but not directly demonstrated in this organism.

Proposed Follow-up Experiments / Actions

1. **Biochemical characterization of purified Q88LM4.** Express and purify the *P. putida* RNase E catalytic domain; measure Mg^{2+} dependence, 5′-monophosphate stimulation, A/U-site specificity, and Zn-link tetramerization to confirm mechanism directly in this organism.
2. **Transcriptome-wide substrate mapping.** Apply RNA-seq/degradome-seq or TIER-seq to an rne conditional depletion strain of KT2440 to catalog direct cleavage sites and define the RNase E regulon, distinguishing it from the RNase G (rng) regulon.
3. **Degradosome pull-down / interactomics.** Co-immunoprecipitation or proximity labeling of tagged RNase E in KT2440 to establish the native degradosome composition (PNPase, RhlB/RhlE helicase, enolase or alternatives) and validate the SLiM-based CTD assembly inferred from *P. aeruginosa*.

4. **CTD truncation genetics.** Construct CTD-deletion and membrane-motif mutants to test their effects on growth, cold/oxidative-stress adaptation, motility, and degradosome assembly — paralleling the *P. aeruginosa* CTD-mutant phenotypes.
5. **Localization imaging.** Fluorescent-fusion single-molecule imaging in live *P. putida* to confirm inner-membrane localization and quantify how the membrane motif and CTD govern diffusion and clustering.
6. **Autoregulation test.** Assay whether the rne 5′UTR forms the conserved stem-loop sensor and mediates feedback control of RNase E levels in KT2440, as in *E. coli*.

Conclusion

The gene **rne / PP_1905 (Q88LM4)** of *Pseudomonas putida* KT2440 encodes the **full-length Ribonuclease E**, the master endoribonuclease of bacterial RNA metabolism. It is a Mg^{2+}-dependent, single-strand-specific endonuclease (EC 3.1.26.12) that cleaves A/U-rich RNA, preferentially at 5′-monophosphorylated substrates via a 5′→3′ scanning mechanism. It matures 16S/5S rRNA and most tRNAs and initiates (usually rate-limits) the decay of most mRNAs. Its zinc-linked catalytic domain forms a homotetramer, while its large C-terminal intrinsically disordered domain scaffolds the RNA degradosome and tethers the machine to the cytoplasmic face of the inner membrane. Direct evidence in the target organism confirms non-redundant, species-specific physiological roles, distinct from its separately encoded paralog RNase G. The UniProt "RNase G subfamily" tag is an automated-annotation artifact superseded by the protein's unambiguous full-length RNase E architecture.
